SARIMAX uses an exogenous regression structure with RMSE 9.209, MAE 7.722, and MAPE 93.356. The model uses an ARMA order of (1, 0, 1) with an intercept and the same lagged exogenous variables used in the preprocessing step. LSTM uses one recurrent layer with dropout and early stopping, and its metrics are RMSE 6.865, MAE 5.394, and MAPE 36.941.

The comparison is intentionally transparent: the SARIMAX baseline is stronger in this phase-one setup, while the LSTM remains a supplementary benchmark. Residual diagnostics are reported for SARIMAX with a residual mean of -7.722 and a first-order autocorrelation of 0.983. The forecast artefacts are saved in the figures and outputs directories, so the project remains easy to reproduce and review for the submission package.

HCP Linkage Memo

Two HCP indicators were examined: FAO_CP_23012 and FAO_CP_23013. A simple regression was used to assess the association between these human-capital indicators and food inflation. The coefficient for FAO_CP_23012 is -0.402 with p-value 0.000, while the coefficient for FAO_CP_23013 is -0.486 with p-value 0.000. These fitted relationships were then used to create forward projections that connect the phase-one inflation forecast to a simple path for human-capital pressure indicators.